

1- Consider the following velocity distribution:

$$\vec{v} = (2x^2 + 2y^2) \hat{x} - 4xy \hat{y}$$

- Is the flow rotational or irrotational?
- Calculate the circulation around the following triangle in a counterclockwise direction.



3.26. Given an incompressible, steady flow, where the velocity is

$$\vec{V} = (x^2y - xy^2)\hat{i} + \left(\frac{y^3}{3} - xy^2\right)\hat{j}$$

- Does the velocity field satisfy the continuity equation? Does a stream function exist? If a stream function exists, what is it?
- Does a potential function exist? If a potential function exists, what is it?
- For the region shown in Fig. P3.26, evaluate

$$\iint (\nabla \times \vec{V}) \cdot \hat{n} dA = ?$$

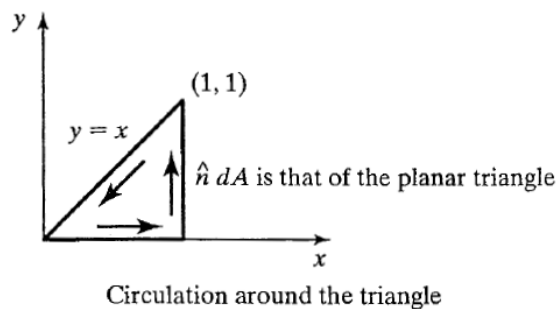


Figure P3.26